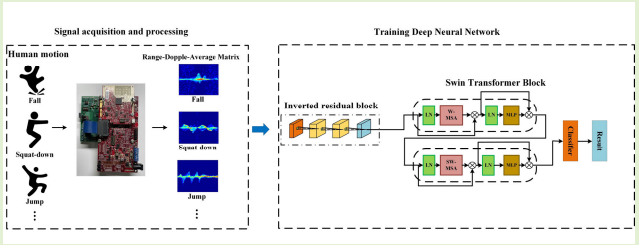
IR-ST: A Lightweight Transformer Network for Human Fall Detection Based on FMCW Radar

Minming Gu, Zhixiang Chen, Kaiyu Chen, and Haipeng Pan

***Abstract*—Without timely treatment, falls often cause serious injuries in elderly individuals. Due to the aging of the global population, an efficient and privacy preserving fall detection system is exceptionally indispensable. According to the above requirements, this article proposes a lightweight neural network inverted residual module and Swin-Transformer module (IR-ST) based on the combination of inverted residual module and Swin-Transformer block. First, the original data are collected by a millimeter-wave radar, and the range-Doppler-average matrix (RDAM) method is used to extract the micro-Doppler map. Second, the spatial feature information in the radar micro-Doppler map is extracted using the inverted residual module. The self-attention mechanism in the Swin-Transformer module is used to learn timing feature information. Finally, the classification results are generated and displayed after processing the data through the fully connected layer. The experiments indicate that the IR-ST neural network exhibits superior accuracy in fall detection, with optimized model parameters that result in reduced prediction time.**



***Index Terms*—Fall detection, frequency-modulated continuous wave (FMCW) radar, inverted residual module and Swin-Transformer module (IR-ST) network, lightweight network, Swin-Transformer block.**

I. Introduction

ELDERLY individuals over 65 years of age will account for one-sixth of the world's total population in 2050 [1]. In the context of an increasingly aging global population, the number of falls among the elderly individuals increases exponentially with age [2]. Therefore, in daily life, it is essential to detect and accurately identify the behavior of elderly individuals.

Optical sensors are used in many applications, such as fall detection and action recognition, to collect images and videos with information about human motion. These images and videos are then analyzed using deep learning methods, which are due to their superior imaging quality [3], [4]. However, this type of sensor still has certain limitations that cannot be avoided, one of which is the requirement of strict environmental conditions. The presence of smoke and other low visibility environments will greatly affect the final imaging effect. Furthermore, it cannot be placed in private areas, such as bathrooms and bedrooms for privacy reasons. Radar sensors can solve the above problems well. By receiving and analyzing echoes related to target characteristics, radar can generally be operated in environments with poor visibility without privacy concerns.

When a target is detected by radar, the motion of the object can cause modulation in the frequency of the radar return signal, resulting in the micro-Doppler effect [5], [6], [7], [8], [9]. Each human movement corresponds to a unique micro-Doppler effect, thus making it possible to detect human movement using this effect. In the past, various body parts were analyzed along with machine learning techniques, such as K-nearest neighbor (KNN) and support vector machine (SVM), to detect human movement [10]. With the rapid development of deep learning technology, a popular approach is to obtain a micro-Doppler map by processing a radar echo signal and using a convolutional neural network (CNN) to classify it. CNN judges the action by determining the characteristic values in the image. Sadreazami et al. [11] used iconography combined with a CNN for training. The accuracy of the five types of action classification was 98.37%. Chen et al. [12] trained 11 types of action datasets through the MobileNetV3 network, and an accuracy of 98.4% was obtained. However, the radar echo signal has prominent timing characteristics, and most studies use long short-term memory (LSTM) networks or LSTM + CNN. The LSTM network can record the relationship between current and historical inputs. It can use the timing characteristics of radar [13], dramatically improving accuracy

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and reducing the model size. Wang et al. [14] extracted the column vector in the radar-Doppler map and trained it through a two-layer LSTM network. The classification accuracy of the six actions reached 92.65% [14]. Liang and Xu [15] extracted the column vectors in the range-Doppler map and used a bi-directional LSTM (Bi-LSTM) network for training, and the accuracy of the four action classifications was 99%. Ma et al. [16] used a CNN and LSTM fusion network to extract the feature information in the Doppler map through CNN and used the timing information through LSTM. Based on the classification of six actions, the accuracy rate was 95%. However, LSTM cannot be used to perform parallel operations. When the time-series data processed is long, a large amount of training and a long training time are required.

The problem of LSTM can be effectively resolved by utilizing the self-attention mechanism present in the transformer [17]. The self-attention module can perform similarity calculations on any pixel point in the radar micro-Doppler image with other pixels. The parallel operation of data can be realized using the timing characteristics in the image and through the self-attention mechanism. The vision transformer (VIT) [18] has leveraged transformers to enhance image processing. VIT has gained substantial recognition in different areas, such as image captioning [19] and safe driving [20], mainly due to its exceptional performance. Zhao et al. [21] used VIT to train the range-doppler map measured by ultrawideband (UWB) radar, and they demonstrated the feasibility of using the self-attention mechanism in the field of radar detection of human motion, with an accuracy of 76.5% in nine activity classifications. The VIT model requires excessive computations for high-resolution images and lacks accuracy in some samples with limited data. To tackle this, Liu et al. [22] proposed the Swin Transformer, which introduces a sliding self-attention mechanism to improve accuracy while reducing computation [23], [24], [25]. However, the Swin-Transformer model is large and not suitable for deployment on devices with limited computing power and memory. Hence, the Swin-Transformer architecture should be optimized to fit radar spectrum data.

In view of the above problems, this article proposes a lightweight neural network composed of inverted residual module and Swin-Transformer module (IR-ST) for human action classification and recognition.

The main contributions of this article are as follows.

1) Range-Doppler-average matrix (RDAM) algorithm is used to process the original Radar intermediate frequency signal, so as to reduce the noise in the complex environment and get a purer radar micro-Doppler graph.
2) The inverted residual module is used to replace the ordinary CNN module to reduce the number of parameters and calculation of the model.
3) Self-attention mechanism is added into the network model, which not only makes up for the problems of long computing time and inability of parallel operation in LSTM network but also has smaller model size and fewer model parameters compared with the existing transformer network.

Through the performance test of the proposed method on the self-built dataset, the results show that the accuracy, model parameters, and prediction time of the proposed method are greatly improved.

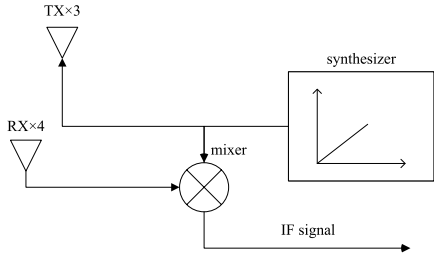


Fig. 1. Millimeter-wave radar signal transmission and reception schematics.

The rest of this article is organized as follows. Section II describes the working principle of millimeter-wave radar and data acquisition and processing methods. Section III introduces the topic of the IR-ST neural network structure and its inverted residual layer structure, self-attention, multihead self-attention (MSA) mechanism module, and Swin-Transformer block layer. Section IV introduces the experimental environment, data collection, and processing, and the network training effect is compared with other networks. Finally, the conclusions drawn from this study are presented in Section V.

II. Signal Collecting and Processing

A. Millimeter-Wave Radar Signal Acquisition

The simplified transceiver module of the FMCW radar system is shown in Fig. 1. The radar used in this experiment has three transmitting antennas and four receiving antennas.

The radar generates a linear frequency-modulated signal through a synthesizer that is emitted by a transmitting antenna (TX). The frequency variation of the signal in a chirp period is as follows [26]:

$$X_{\mathrm{TX}}(\mathrm{t}) = \cos\left(2\pi f_c t + \pi\beta t^2\right) \tag{1}$$

where f_c is the chirp's starting frequency. β is the chirp's slope, which can be expressed by

$$\beta = \frac{B_{\mathrm{chirp}}}{T_{\mathrm{chirp}}} \tag{2}$$

where B_{chirp} is the chirp bandwidth and T_{chirp} is the chirp duration. When the transmitted signal is reflected by the environment and the target, it is received by the receiving antenna (RX). The received signal can be represented by

$$X_{\mathrm{RX}}(\mathrm{t}) = \alpha\cos\left(2\pi f_c(t-\tau) + \pi\beta(t-\tau)^2\right). \tag{3}$$

In (3), considering factors such as atmospheric attenuation and target scattering, α is the attenuation coefficient of radar received signal amplitude and τ is the time delay

$$\tau = \frac{2\mathrm{d}}{\mathrm{c}} \tag{4}$$

where d is the distance between the radar and the object and c is the speed of light.

After the signal has been mixed via the mixer, the resulting radar IF signal is obtained, which can be represented by the following:

$$\begin{aligned} X_{\mathrm{TR}}(\mathrm{t}) &= X_{\mathrm{TX}}(\mathrm{t})X_{\mathrm{RX}}(\mathrm{t}) \\ &= \cos\left(2\pi\beta\tau\mathrm{t} + 2\pi f_c\tau - \pi\beta\tau^2\right). \end{aligned} \tag{5}$$

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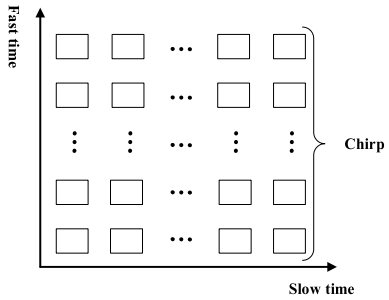


Fig. 2. Processing the radar Doppler matrix using the RDM algorithm.

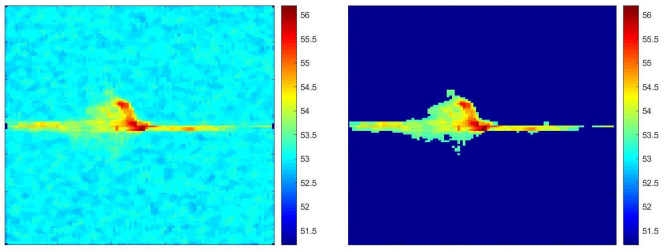


Fig. 3. Representative set of radar micro-Doppler images of fall action is extracted. (a) Effect of RDM algorithm processing. (b) Effect of RDAM algorithm processing.

B. RDAM Signal Processing

The radar mixed IF signal obtained by the above method can be processed by the RDAM algorithm to obtain the radar range-Doppler graph. The RDAM algorithm is composed of two parts: the range-Doppler matrix (RDM) algorithm and average noise reduction algorithm. The RDM algorithm is responsible for creating the micro-Doppler matrix, while the average noise reduction algorithm helps to reduce the noise and achieve a clearer radar image.

The RDM algorithm can be divided into two parts: range fast Fourier transform (FFT) and Doppler FFT. As shown in Fig. 2, the ordinate is the fast time channel, which is composed of a single-cycle chirp. Because each frequency sweep cycle is very short, the speed information in each chirp can be ignored. Range FFT refers to extract the abscissa frequency corresponding to the spectrum peak value for each chirp on the fast time channel to obtain the distance information of the target. The distance calculation (6) is as follows:

$$R = \frac{cT}{2f} \cdot F \tag{6}$$

where T is the sweep period, f is the sweep bandwidth, and F is the frequency of the radar beat signal in motion or static conditions.

The horizontal coordinate is the slow time channel, which consists of several groups of chirp cycles. In the slow time channel, the speed information of the measured object is included. Doppler FFT is used to calculate the frequency of the data on the slow time channel. The speed calculation in (7) is as follows:

$$V = \frac{f_d c}{2f} \tag{7}$$

where f_d is the Doppler frequency. The RDM algorithm is used to rearrange the mixing signal to obtain the micro-Doppler matrix.

The collected micro-Doppler graphs exhibit high levels of environmental disturbances. This article employs the average noise reduction method to filter environmental clutter. The method involves three steps: 1) determining the average value of all data in the radar micro-Doppler matrix; 2) comparing each data with the average value and retaining the data that exceeds the average value while deleting those that are below the average value; and 3) averaging the remaining data and repeating Step 2. This process results in a cleaner radar micro-Doppler map.

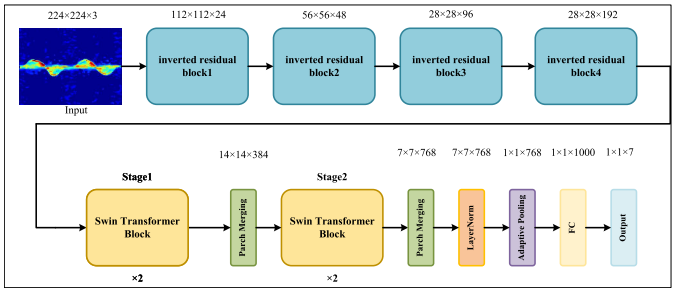


Fig. 4. IR-ST neural network flowchart.

As shown in Fig. 3, the RDAM algorithm can be used to filter out environmental interference and highlight the principal eigenvalues.

III. Neural Network Model Design

A. Neural Network Model Diagram

Based on inverted residual module and Swin-Transformer block, a lightweight CNN + Transformer neural network named IR-ST network is proposed. The structure diagram of IR-ST is shown in Fig. 4.

The algorithm begins by extracting a radar micro-Doppler image using four inverted residual modules. The size of the input matrix is reduced by half in the one to three layers inverted residual module, while the depth is doubled. In the fourth layer, the depth of the matrix is doubled without affecting the size of the input matrix. Next, the input matrix is subjected to self-attention mechanism operation by two Swin-Transformer blocks. After the operation, the matrix size is reduced to 14×14 and 7×7 by the parch merging structure, and then, the matrix data are normalized by the LayerNorm (LN) layer. Adaptive pooling is then utilized to reduce the matrix size to 1×1. Finally, the algorithm produces classification results by passing through the full connection layer.

B. Inverted Residual Structure

CNN networks are excellent at extracting feature information from images. However, as the network becomes deeper, the model size and operation time increase. The Google team proposes an inverted residual structure that is based on convolution [27]. The inverted residual structure provides an efficient solution by combining pointwise (PW) and depthwise (DW) convolutions to replace traditional convolution layers. This results in a significant reduction in the model size of the network.

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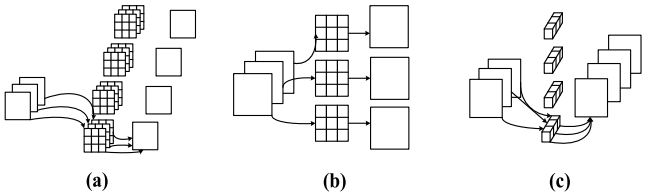


Fig. 5. Three different convolution methods. (a) Ordinary convolution. (b) DW convolution. (c) PW convolution.

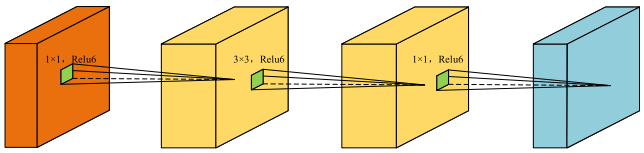


Fig. 6. Inverted residual block.

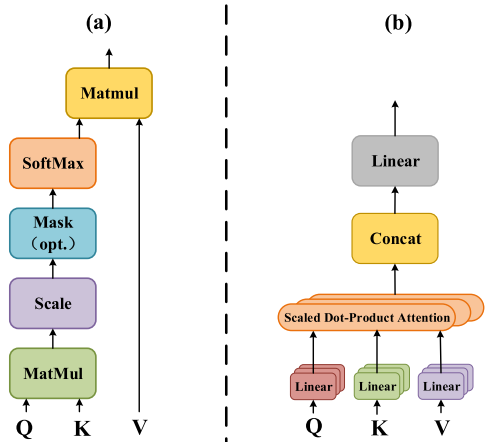


Fig. 7. Two self-attention mechanism schematics. (a) Self-attention. (b) Multihead attention.

According to Fig. 5(a), traditional convolution techniques involve performing a convolution operation on each channel of the input matrix using a set of convolution kernels. The resulting processed convolution feature matrix is then combined to create a channel, and this operation is repeated to form a multichannel matrix that serves as an input for the next layer in the network. The amount of calculation for ordinary convolution can be derived from the following equation:

$$S_O = D_K \cdot D_K \cdot A \cdot C \cdot H \cdot W \tag{8}$$

where D_K is the size of the convolution kernel, H is the height of the input matrix while W represents its width, C is the number of channels of the input matrix, and A is the number of convolution kernels.

As shown in Fig. 5(b) and (c), in DW convolution, only convolution operations are performed on each channel, and in PW convolution, a set of 1×1 convolution kernel input matrices are used to increase the dimension. The amount of calculation for the DW and PW convolutions can be obtained as follows:

$$S_D = D_K \cdot D_K \cdot C \cdot H \cdot W \tag{9}$$

$$S_P = A \cdot C \cdot H \cdot W. \tag{10}$$

In this study, the DW and PW convolution operations are combined to construct the inverted residual structure, and the feature extraction of the radar range-Doppler map is carried out using this structure. As shown in Fig. 6, the depth of the input matrix is first increased by three times through PW convolution. Then, feature extraction is performed by DW convolution with a convolution kernel size of 3×3 and a step distance of 2. Finally, the depth of the input matrix is reduced to 2/3 of the original using PW convolution.

In this article, the ratio of the amount of calculation of the inverted residual structure constructed by DW + PW convolution to the amount of calculation of ordinary convolution is as follows:

$$\frac{S_{P1} + S_D + S_{P2}}{S_O} = \frac{3 \cdot C \cdot C \cdot H \cdot W + D_K \cdot D_K \cdot 3 \cdot C \cdot H \cdot W + 2C \cdot 3 \cdot C \cdot H \cdot W}{D_K \cdot D_K \cdot 2 \cdot C \cdot C \cdot H \cdot W}$$

$$= \frac{1}{6} + \frac{3}{2 \cdot C} + \frac{1}{3}. \tag{11}$$

When the number of channels of the input matrix is 24, 48, and 96, using the inverted residual structure can reduce the amount of calculation by half in each convolution operation.

C. Swin-Transformer Block

To solve the problem of the limited receptive field of CNNs, a self-attention mechanism [13] is proposed. As shown in Fig. 7(a), the self-attention mechanism is used to construct three trainable weight matrices: W^Q, W^K, and W^V. The input sequence a is calculated by (12) to obtain the matching item Q, the key item K, and the information V in the input sequence

$$Q = \mathrm{a} \cdot \mathrm{W}^Q, K = \mathrm{a} \cdot \mathrm{W}^K, V = \mathrm{a} \cdot \mathrm{W}^V. \tag{12}$$

Finally, the self-attention mechanism calculation can be expressed by the following formula:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{13}$$

where d_k refers to the number of elements in K. $(QK^T/(d_k)^{1/2})$ is calculated based on the matrix, normalized by the softmax function, so that the sum of the elements is equal to one and finally multiplied by the information V extracted from the input of the self-attention mechanism.

In some complex environments, the use of a single-head self-attention mechanism will be limited by the feature space, so an MSA mechanism is introduced. Fig. 7(b) is a flowchart of an MSA mechanism. It has multiple self-attention mechanism channels, and a self-attention calculation is performed in each channel. The formulas are given as follows:

$$Q_\mathrm{i} = \mathrm{a} \cdot \mathrm{W}_\mathrm{i}^Q, \quad K_\mathrm{i} = \mathrm{a} \cdot \mathrm{W}_\mathrm{i}^K, \quad V_\mathrm{i} = \mathrm{a} \cdot \mathrm{W}_\mathrm{i}^V \tag{14}$$

$$I_\mathrm{i} = \mathrm{Attention}(Q_\mathrm{i}, K_\mathrm{i}, V_\mathrm{i}) \tag{15}$$

$$\mathrm{Multi\text{-}Head}(Q, K, V) = \mathrm{Concat}(I_1, I_2, I_3, \ldots) \tag{16}$$

where W_i^Q, W_i^K, and W_i^V are the learnable training matrices in the ith self-attention mechanism channel. First, Q_i, K_i, and V_i are calculated by (14). Second, the output vector in the ith

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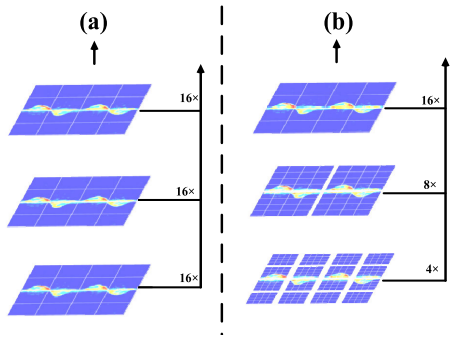


Fig. 8. Sampling methods of different neural networks. (a) Vision transformer. (b) Swin transformer.

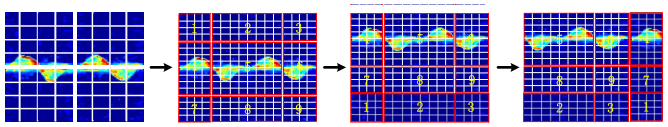


Fig. 9. Sliding window principal diagram.

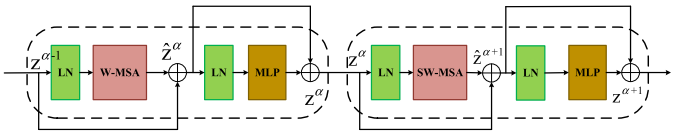


Fig. 10. Swin-Transformer block schematic.

channel is calculated by (15). Finally, the output vectors of each layer are spliced. Using the multihead attention mechanism, the correlation between features can be extracted from multiple angles without increasing the amount of computation, thereby obtaining richer feature information and improving training accuracy.

The VIT [14] approach partitions the image into sections and introduces position embedding to train the transformer encoder. Swin Transformer, on the other hand, adopts a CNN architecture based on VIT, introducing a window attention mechanism module while maintaining the hierarchical structure. Fig. 8(a) demonstrates the attention mechanism of the VIT network. It depicts that the downsampling rate remains constant as the network depth grows. On the contrary, Fig. 8(b) shows how the Swin-Transformer block splits the image into nine smaller modules followed by the transformer operation. Each module utilizes a separate attention mechanism. As the network gets deeper, the small modules are combined to create larger modules, leading to an increase in the sampling rate.

When using windows MSA (W-MSA), there is no attention mechanism between the windows. As shown in Fig. 9, the Swin Transformer uses a sliding window that allows windows to communicate with each other. First, the image is segmented by nine windows. Second, to ensure that the size of each window is the same, windows 1–3 are first moved down, and then, windows 1, 4, and 7 are moved right. Finally, windows 2, 4, 6, and 8 and windows 1, 3, 7, and 9 are spliced. The output result is consistent with the size of the first image, thus realizing the information interaction between different windows.

The Swin-Transformer block consists of two modules, as shown in Fig. 10. Each module has four parts, LN, MSA module, and multilayer perceptron (MLP) module. The MSA module is divided into a W-MSA module and a sliding window-based self-attention machine (SW-MSA) module. The Swin-Transformer block formulas are given as follows:

$$\hat{Z}^r = \text{W-MSA}(\text{LN}(Z^{r-1})) + Z^{r-1} \tag{17}$$

$$Z^r = \text{MLP}(\text{LN}(\hat{Z}^r)) + \hat{Z}^r \tag{18}$$

$$\hat{Z}^{r+1} = \text{SW-MSA}(\text{LN}(Z^r)) + Z^r \tag{19}$$

$$Z^{r+1} = \text{MLP}(\text{LN}(\hat{Z}^{r+1})) + \hat{Z}^{r+1} \tag{20}$$

where $\hat{Z}^r$, Z^r(Z^{r+1}), and $\hat{Z}^{r+1}$ represent the output matrix of module r after going through W-MSA, MLP, and SW-MSA modules, respectively.

Formula (21) is the calculation amount of the VIT attention module, and (22) is the calculation amount of the Swin-Transformer window attention module, where Ω represents the amount of computation for two different structures of self-attention mechanisms. M is the size of the window. When the input matrix size H $\times$ W is much larger than the window size M, the use of W-MSA + SW-MSA remarkably reduces the calculation amount of the network

$$\Omega(\text{MSA}) = 4\text{HWC}^2 + 2(\text{HW})^2 C \tag{21}$$

$$\Omega(\text{W-MSA} + \text{SW-MSA}) = 2(4\text{HWC}^2 + 2M^2\text{HWC}). \tag{22}$$

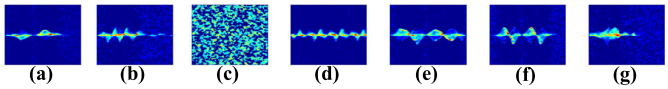


Fig. 11. Radar range-Doppler images of different human actions processed by RDAM algorithm. (a) Sitting up. (b) Jumping. (c) Resting. (d) Marking time. (e) Squatting down. (f) Stooping. (g) Falling.

IV. Experimental Design and Result

A. Dataset Construction

This experiment utilizes the IWR6843isk radar module, developed by Texas Instruments, to gather raw data. The radar operates in a frequency band of 60–64 GHz. Fig. 11 shows the range-Doppler map of seven actions obtained by the RDAM algorithm.

In order to make the data have good generalization ability, a total of 12 experimenters with different ages, heights, and weights participated in the experiment and built the dataset. The parameters of age, height, and weight of 12 experimenters are shown in Table I. In addition to falling, the experiment measured rest, jumping, timing, sitting up, stopping, and squatting six movements. Each action has 300–600 pictures and a total of 3279 sets of data. The amount of data for each action is shown in Table II, 80% of which is the training set and 20% is the test set. The experiment was carried out in an office environment, as depicted in Fig. 12, to demonstrate the method's anti-interference ability. During the experiment, the office personnel acted normally while following the actions as described in the experiment. The items were placed in a usual manner.

B. Network Training Environment and Parameters

In this article, the IR-ST network is created. To prevent overfitting during training, the input image size of the training

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Fig. 12. Laboratory environment when collecting raw data.

TABLE I
AGE, HEIGHT, AND WEIGHT DATA OF 12 SUBJECTS

Tester	Age	Height(cm)	Weight (kg)
1	21	180	80
2	23	170	57
3	24	168	68
4	24	174	74
5	24	181	62
6	26	179	80
7	40	173	64
8	42	152	57
9	43	168	70
10	49	155	55
11	51	173	90
12	55	160	70

TABLE II
NUMBER OF ACTIONS IN THE RADAR ACTION DETECTION DATASET 7

Human motion	Quantity
Resting	483
Falling	500
Jumping	432
Stooping	379
Squatting down	424
Marking time	552
Sitting up	509

set is randomly cropped to 224 × 224 by the algorithm. The data processing and training platform is Windows 10, which has an Intel i7 CPU and NVIDIA3080Ti GPU. Moreover, the neural network environment is Python 3.7. Using the AdamW optimizer, the learning rate is set to 0.0001.

C. Evaluation Indicators

Accuracy, precision, recall, and specificity are used to evaluate the training effect of the network. Accuracy refers to the proportion of actions that are correctly predicted based on all actions, precision refers to the proportion of actions that are correctly predicted to be the same action as other actions in the predicted situation, recall refers to the proportion of actions that are correctly classified in the actual situation, and specificity refers to the proportion of actions that are judged not to be the same action in an image that does not belong to a specific action. The above index calculation formula is as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{23}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{24}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{25}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{26}$$

where TP means true positive, FP means false positive, TN means true negative, and FP means false positive.

TABLE III
EFFECT OF IR-ST NETWORK ON SEVEN KINDS OF MOVEMENT TRAINING

Human motion	Precision	Recall	Specificity
Resting	0.990	1.0	0.998
Falling	0.980	1.0	0.996
Jumping	1.0	1.0	1.0
Stooping	1.0	0.987	1.0
Squatting down	0.976	0.988	0.996
Marking time	1.0	1.0	1.0
Sitting up	1.0	0.970	1.0

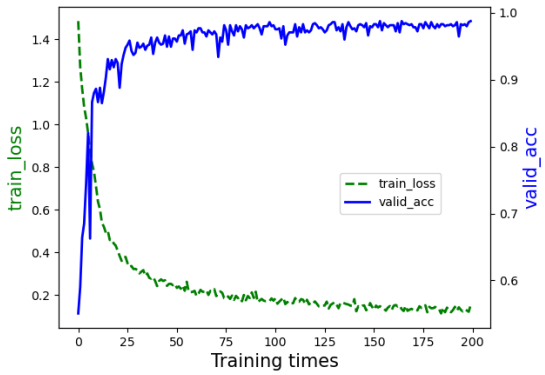


Fig. 13. IR-ST network training set error decrease chart and test set accuracy rate increase chart.

D. Experimental Results

To explore the improvement of the IR-ST network in fall detection, this article first analyzes the training indicators of the IR-ST network itself. Furthermore, the four parameters of accuracy, model size, model parameters, and prediction time are compared with other commonly used neural networks and IR-ST networks.

Fig. 13 displays the loss of the training set and the accuracy of the test set. In the figure, the green dashed line is the loss curve of the training set, and the blue solid line is the accuracy curve of the test set. It indicates a clear trend of convergence in terms of test set error. The accuracy of the test set demonstrates a significant upward trend as the quantity of training increases, culminating in a peak of 99.2%.

Fig. 14 shows the optimal confusion matrix obtained after five cross-validations. It shows the number of correct numbers per action determined by the actual label (horizontal) and the

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TABLE IV
TRAINING ACCURACY OF NINE NEURAL NETWORKS

	IR-ST	MobileNet V2	MobileNet V3-s	MobileNet V3-l	Three-layer LSTM	AlexNet	VGG16	VIT-B/16	Swin-T
Resting	0.990	0.985	0.971	0.990	0.995	0.995	0.995	0.995	0.990
Falling	0.980	0.995	0.830	0.974	0.971	0.995	**1.0**	0.880	**1.0**
Jumping	**1.0**	0.971	0.806	0.955	0.976	0.966	**1.0**	0.959	0.988
Stooping	**1.0**	0.880	0.909	0.892	0.729	0.953	0.949	0.847	0.831
Squatting down	0.976	0.979	0.870	0.954	0.902	0.953	0.988	0.806	0.901
Marking time	**1.0**	0.969	0.972	0.991	1.0	**1.0**	**1.0**	0.995	0.982
Sitting up	**1.0**	0.957	**1.0**	0.953	0.918	0.985	0.981	0.860	0.969
Avg	**0.992**	0.963	0.914	0.961	0.930	0.980	0.989	0.910	0.956

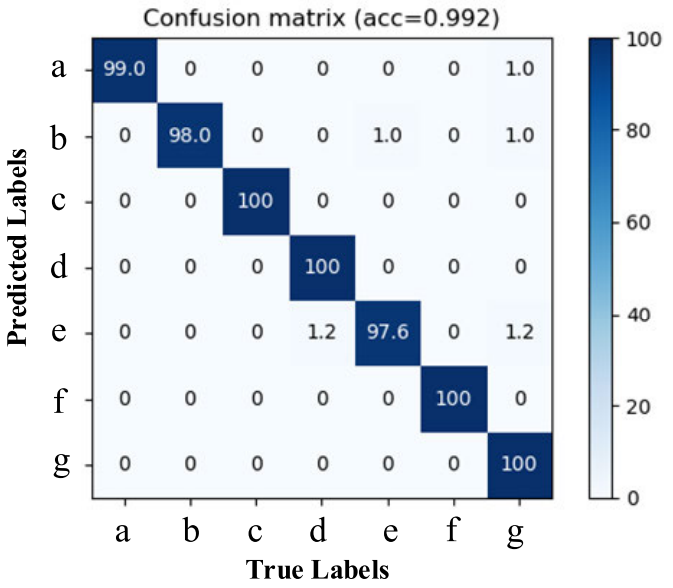


Fig. 14. Confusion matrix of the IR-ST network test set, where (a) rest, (b) fall, (c) jump, (d) stoop, (e) squat down, (f) marking time, and (g) sitting up.

prediction label (vertical). The matrix indicates that out of 652 radar micro-Doppler images, only five predictions were incorrect. The accuracy, recall, and specificity of each action are shown in Table III. The three indicators for falls were 0.980, 1.0, and 0.998, respectively. This indicates that only two groups of data related to fall movements were wrongly classified as squatting down and sitting up, respectively. None of the other movements were misjudged as falls. Therefore, the accuracy requirements for fall detection have been fulfilled.

Table IV presents a comparison of the accuracy for eight neural networks in predicting the actions of seven distinct groups, as well as the average predictive accuracy across all groups. The findings illustrate that the IR-ST network, which is proposed in this article, has the highest recognition accuracy at 99.2%. When using IR-ST, there is an improvement of 2%–8% in accuracy compared to the three-layer LSTM and MobileNet networks, and this clearly shows the advantages of using IR-ST in lightweight networks.

Table V shows a comparison of model sizes, parameters, and average forecast time per image for different networks. Compared with the VGG16 network with similar accuracy, the model size is reduced by 81×. Compared with three-layer LSTM and MobileNet networks, the IR-ST network model size has been slightly reduced. Moreover, the

TABLE V
SIZE, PARAMETERS, AND AVERAGE PREDICTION TIME FOR NINE NEURAL NETWORKS

Network	Size (MB)	Params (M)	Time (ms)
IR-ST	6.3	**2.0**	21.9
MobileNet V2	8.73	3.5	27.5
MobileNet V3-s	**5.94**	2.5	**21**
MobileNet V3-l	16.2	5.4	44
Three-layer LSTM	29	7.62	28.5
AlexNet	55.6	16.63	47
VGG16	512	138.35	372
VIT-B/16	329	103.1	635
Swin-T	105	28	193

model parameters of the IR-ST network are the least among the nine networks, and the average prediction time for a single image only requires 21.9 ms, which meets the real-time prediction requirements. Therefore, the IR-ST network has the advantages of high accuracy, fast prediction, and small size.

V. CONCLUSION

In this article, a fall detection algorithm is proposed that combines micro-Doppler map and an IR-ST network. The network utilizes inverted residual modules and Swin-Transformer blocks to extract the features of the radar micro-Doppler map and incorporate temporal information in the graph for improved detection performance and network lightweight ness. Compared to other network structures, the IR-ST network structure demonstrated better accuracy and weightlessness. Experimental results showed significant improvement in detection accuracy, reduced computational delay, and improved generalization ability in multiaction detection. Future work includes designing a real-time fall detection system using embedded devices like Raspberry Pi.

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